

Design of Traction Substation Based on BIM Technology

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Abstract—This paper explores the application of Building Information Modeling technology in the design of traction substations through a practical project case. Utilizing Autodesk Revit software, a three-dimensional model of the traction substation is developed, encompassing the creation of an electrical equipment family library and the generation of a detailed 3D layout model. The study further elaborates on the application of Building Information Modeling technology in model collision detection, detailed design, and visual representation. The results demonstrate that BIM technology significantly enhances the design, construction, management, and operation and maintenance levels of traction substations, highlighting its substantial application value in modern railway infrastructure.

Keywords—Revit; Traction Substation; Parametric Modeling; Coordination Management

I. INTRODUCTION

A. Research topic and background

As the core component of the electrified railway power supply system, the construction quality and operation stability of the traction substation are directly related to the safety and efficiency of the railway operation. With the rapid development of railway construction, especially the high-speed railway, the railway industry has put forward higher requirements for the standardization, refinement and intelligence of survey and design. In this context, the traditional two-dimensional design method has gradually exposed its limitations, and it is difficult to meet the needs of modern railway design. Building information Model (Building Information Modeling, BIM), as an engineering data model based on 3 D digital technology, provides a new solution for the design, construction, operation and maintenance of railway infrastructure [1].

B. Problems in the current design

At present, the design of railway traction substation mainly relies on the traditional two-dimensional design tools (such as AutoCAD), which has obvious deficiencies in spatial visualization, multi-professional collaborative design and optimization of complex cable layout. For example, two-dimensional design is difficult to accurately express the spatial relationship between equipment, which easily leads to design conflicts and construction rework, at the same time, the phenomenon of information isolation between multiple specialties also increases the difficulty of design coordination. These problems not only reduce the design efficiency, but also may affect the construction quality and the project schedule.

C. Challenges and solutions in the research

In the design of traction substations, major challenges include spatial coordination of complex equipment layouts, optimization of cable routing, and integration of multi-disciplinary collaborative workflows. To address these issues, this study introduces BIM technology, leveraging functions such as 3D modeling, parametric design, and collision detection to achieve comprehensive optimization of the design process. BIM technology effectively resolves spatial conflicts in traditional design methods and enhances communication efficiency through visualization. Additionally, the collaborative platform feature of BIM provides a unified working environment for multidisciplinary design teams, significantly improving design coordination [2].

D. Research purpose and method

This study aims to explore the application of BIM technology in the design of traction substation, and to build a digital three-dimensional design platform based on BIM to improve the accuracy, synergy and efficiency of the design. To achieve this goal, this study used Autodesk Revit as a core modeling tool, combined with 3 dsMAX and Unity3D for visual presentation, and verified the advantages of BIM technology in collision detection, cable path optimization, and construction simulation through practical cases. Compared with recent studies, this study not only fills the gap in the application of BIM in the design of railway traction substation, but also verifies its feasibility and effectiveness through practical engineering cases.

E. The significance and novelty of the research

The significance of this study is to introduce BIM technology into the design field of railway traction substation, which provides theoretical support and practical reference for the digital and intelligent transformation of railway infrastructure. Its novelty is reflected in the following aspects: Build the BIM equipment family library suitable for the traction substation, and realize the parametric modeling and rapid call of the equipment.

The BIM-based cable path optimization method is proposed, which significantly improves the design efficiency and construction feasibility.

Through three-dimensional visualization and virtual simulation technology, it provides an intuitive tool for design review and construction disclosure. These innovations not only improve the design quality of traction substations, but also provide new ideas and methods for the digital transformation of

the railway industry [3]. It provides a comprehensive theoretical analysis and practical verification for the application of BIM technology in the design of railway traction substation, and has important academic value and engineering significance.

II. COMPARATIVE ANALYSIS OF TRADITIONAL DESIGN AND BIM TECHNOLOGY

A. Limitations of Traditional Design Methods

Currently, the design of railway electrification systems in China predominantly relies on two-dimensional (2D) software such as AutoCAD. While traditional design methods are well-established, they exhibit several significant limitations that hinder the efficiency and accuracy of modern infrastructure projects[4].

- **Lack of Spatial Visualization:** Traditional 2D design methods struggle to represent complex spatial relationships within traction substations. For instance, intricate arrangements of electrical equipment, cable trays, and structural elements often require multiple views and cross-sections to convey the design intent. This fragmented approach frequently leads to omissions, misinterpretations, and errors that may remain undetected until the construction or operational phases. Such late-stage discoveries not only result in costly rework but also pose significant safety risks, particularly in high-voltage environments where precision is paramount.
- **Cable Layout Challenges:** The design of cable routing in traction substations is a highly complex task due to the dense network of power and control cables. Traditional 2D methods fail to accurately depict the spatial relationships between cables, conduits, and other infrastructure components. This limitation often results in flawed designs, such as cable crossovers, insufficient bending radii, or inadequate clearance distances. These issues are typically identified only during construction, leading to delays, material waste, and increased labor costs. Moreover, suboptimal cable management can compromise the long-term reliability and maintenance efficiency of the substation.
- **Interdisciplinary Coordination Issues:** Traction substation design involves multiple disciplines, including electrical, civil, and mechanical engineering. Traditional design methods lack a unified platform for interdisciplinary collaboration, making it difficult to coordinate interface relationships between disciplines. For example, conflicts between electrical conduits and structural beams or HVAC ducts are often overlooked during the design phase, only to be discovered during construction. This lack of coordination not only reduces the overall design quality but also increases the risk of project delays and budget overruns.

B. Advantages of BIM Technology

Building Information Modeling (BIM) technology addresses the limitations of traditional design methods by creating a comprehensive, data-rich 3D model that integrates all aspects of the project. This transformative approach offers several key

advantages that enhance the design, construction, and operation of traction substations [5].

- **Enhanced Visualization and Communication:** BIM provides a highly realistic and interactive 3D representation of the traction substation, enabling stakeholders to visualize the entire project in detail. This capability facilitates better communication and decision-making among designers, contractors, and clients. For example, 3D models can be used to conduct virtual walkthroughs, allowing stakeholders to identify potential issues and propose solutions before construction begins. This proactive approach minimizes misunderstandings and ensures that all parties are aligned with the project objectives.
- **Advanced Simulation and Analysis Capabilities:** BIM supports a wide range of simulations and analyses that are impossible with traditional 2D methods. For instance, collision detection algorithms can automatically identify clashes between electrical conduits, structural elements, and mechanical systems, enabling designers to resolve conflicts during the design phase. Additionally, BIM tools can simulate insulation distance checks, thermal performance, and electromagnetic interference, ensuring that the design complies with safety and operational standards. These capabilities not only improve design accuracy but also reduce the risk of costly modifications during construction.
- **Improved Interdisciplinary Coordination:** BIM serves as a centralized platform for interdisciplinary collaboration, enabling real-time coordination between electrical, civil, and mechanical disciplines. By integrating all design data into a single model, BIM eliminates the silos that often hinder traditional design processes. For example, electrical engineers can instantly access updates from civil engineers regarding structural changes, ensuring that cable routing and equipment placement are adjusted accordingly. This seamless coordination reduces design conflicts, enhances efficiency, and ensures that the final design is both functional and constructible.
- **Life-cycle Management and Sustainability:**

BIM extends its benefits beyond the design and construction phases, supporting the entire lifecycle of the traction substation. The data-rich BIM model serves as a digital twin of the physical asset, providing valuable information for operation and maintenance (O&M) activities. For instance, facility managers can use the BIM model to locate specific components, track maintenance schedules, and plan upgrades. This capability not only improves the efficiency of O&M activities but also supports sustainability initiatives by optimizing resource utilization and extending the lifespan of the substation.

III. TRACTION SUBSTATION DESIGN BASED ON REVIT PLATFORM

This study presents a detailed methodology for designing a traction substation using Autodesk Revit as the primary BIM tool. The case study focuses on a 110kV traction substation employing a direct power supply mode with a return wire. The 110kV distribution equipment is arranged outdoors, while the

27.5kV distribution system, excluding manual isolation switches, lightning coils, and arresters, utilizes an indoor AIS switchgear layout. The main transformer and terminal box are floor-mounted, whereas other outdoor distribution equipment adopts a pole-type configuration.

A. Software platform selection

The design process leverages Autodesk Revit as the core modeling tool due to its seamless interoperability with CAD software, robust parametric family capabilities, and extensive plugin ecosystem, which collectively enable efficient and precise modeling. For enhanced visualization, 3dsMAX is employed for rendering, and Unity3D is utilized for interactive 3D visualization, providing a comprehensive digital representation of the substation[6].

B. Modeling process

The modeling workflow is a comprehensive and iterative process that begins with a meticulous analysis of the traction substation design drawings. This analysis encompasses the overall layout, equipment specifications, connection methods, and architectural details, ensuring that all aspects of the substation are accurately represented in the digital model. The process involves the following detailed steps:

- **Data Collection and Preparation:** Extract and summarize key information from 2D drawings, including equipment types, dimensions, and spatial relationships.
- **Equipment Family Library Development:** Create parametric models of critical components, such as transformers, circuit breakers, and disconnectors, based on manufacturer specifications and design requirements.
- **Substation Layout Modeling:** Construct the architectural framework, including the production house, yard, and perimeter walls, using Revit's architectural tools.
- **Equipment Placement and Integration:** Position the equipment models within the architectural layout and establish electrical connections, ensuring compliance with design standards and spatial constraints.
- **Model Enhancement and Visualization:** Export the completed BIM model to 3dsMAX for high-quality rendering and to Unity3D for interactive visualization, enabling stakeholders to explore the substation in a virtual environment.

The detailed modeling workflow is illustrated in Figure 1.

C. Establishment of equipment family library

The traction substation comprises a wide range of critical equipment, including transformers, circuit breakers, disconnectors, voltage transformers, current transformers, and AIS switchgear. The creation of an equipment family library is a fundamental step in the BIM-based design process, ensuring that all components are accurately represented and can be efficiently reused across multiple projects. The development of this library involves the following detailed steps:

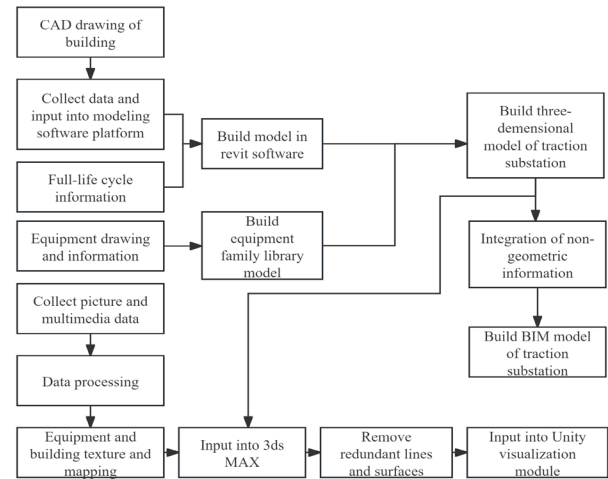


Fig. 1. Modeling workflow

- **Data Collection:** The first step involves gathering comprehensive geometric and non-geometric data for each piece of equipment. Geometric data includes dimensions, shapes, and spatial configurations, while non-geometric data encompasses electrical parameters (e.g., voltage ratings, current capacities), material specifications, and operational constraints. This information is typically sourced from equipment manufacturers' datasheets, design documents, and industry standards.
- **Parametric Modeling:** Using Autodesk Revit's parametric modeling tools, adjustable 3D models of each equipment type are developed. Parametric modeling allows designers to define constraints and relationships between components, ensuring that the models are flexible and adaptable to different design scenarios. For example, a transformer model can be adjusted to accommodate varying power ratings or physical dimensions by modifying a few key parameters. This approach not only enhances design efficiency but also ensures consistency and accuracy across the project.
- **Validation and Optimization:** Once the initial models are created, they are rigorously validated against real-world data to ensure accuracy and functionality. This process involves cross-referencing the models with manufacturer specifications, conducting virtual tests (e.g., insulation distance checks), and simulating operational scenarios. Any discrepancies or inefficiencies are identified and addressed during this phase. Additionally, the models are optimized for reuse in future projects by standardizing parameters and ensuring compatibility with other BIM components. This optimization enhances the long-term value of the library, reducing the time and effort required for subsequent designs [7].

Examples of the traction transformer and circuit breaker models are shown in Figures 2 and 3, respectively.

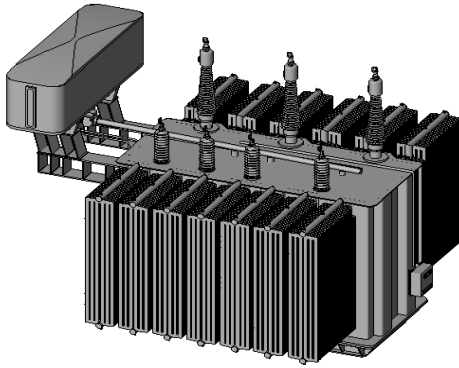


Fig. 2. Traction transformer model

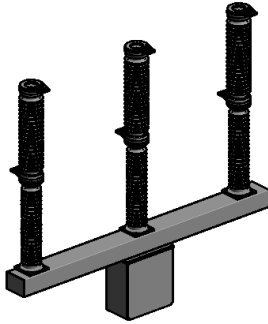


Fig. 3. Circuit breaker model

D. Traction substation building and general layout

Following the development of the equipment family library, the overall BIM model of the traction substation is constructed. This phase involves a systematic and detailed design process to ensure that all architectural, structural, and electrical components are seamlessly integrated into a cohesive and functional 3D model. The design process is divided into the following key steps:

- **Architectural Framework Development:** Establish the architectural axis network, elevation levels, and coordinate system based on 2D drawings. The axis network serves as the backbone of the model, providing a reference for the placement of all structural and equipment elements. Elevation levels are critical for ensuring that the substation's layout adheres to spatial constraints and operational requirements.
- **Structural and Equipment Integration:** Position the equipment within the architectural framework, ensuring proper alignment and spatial coordination. This involves placing transformers, circuit breakers, disconnections, and other critical components in their designated locations. The BIM model allows for real-time visualization and adjustment, ensuring that all elements are optimally positioned to meet design specifications and operational needs.
- **Electrical System Design:** Complete the wiring and connection of electrical components, adhering to design specifications and safety standards. The BIM model enables precise cable routing, minimizing the risk of crossovers, insufficient bending radii, or inadequate

clearance distances. Additionally, the model supports the simulation of electrical performance, such as load distribution and insulation checks, ensuring that the design meets operational requirements and safety regulations

The final BIM model, including the general layout of the substation and the production house, which is depicted in Figures 4 and 5.

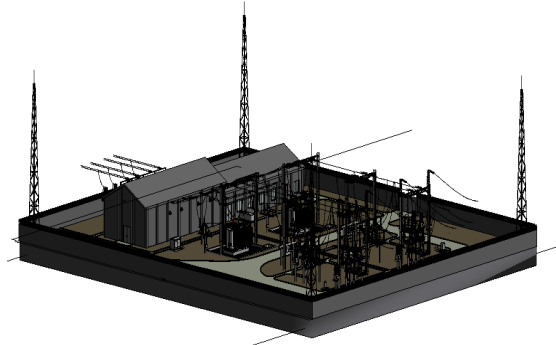


Fig. 4. BIM model of traction substation

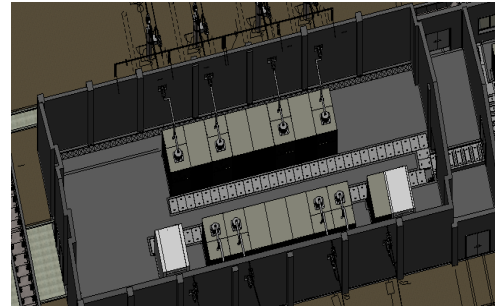


Fig. 5. BIM model of production house

IV. BIM COLLABORATIVE APPLICATIONS IN TRACTION SUBSTATION DESIGN

A. Collision Detection and Resolution

The integration of BIM technology enables comprehensive collision detection by combining mechanical, electrical, and architectural models into a unified digital environment. Utilizing advanced algorithms, BIM software identifies and resolves spatial conflicts between pipelines, cable trays, and structural elements during the design phase. This proactive approach minimizes the risk of "clash, leakage, and omission" issues, which are common in traditional design methods. For instance, this capability not only enhances design accuracy but also reduces rework, delays, and associated costs, ensuring a smoother construction process [8].

B. Cable Routing Optimization

BIM technology facilitates the creation of a detailed 3D cable routing model, which simulates the installation process and optimizes the cable layout. The system automatically detects conflicts between cables and other components, such as structural beams and conduits, and generates an optimized routing path. Additionally, it produces a comprehensive cable laying sequence table, which includes critical information such

as the installation sequence, start and end points, cable specifications, and lengths. This table serves as a practical guide for on-site construction teams, ensuring efficient and accurate cable installation. The optimized cable routing model achieves several key benefits:

- **Material Efficiency:** Reduces cable waste through precise length calculations and path optimization.
- **Improved Aesthetics:** Ensures a clean and organized cable layout, enhancing the overall appearance of the substation.
- **Enhanced Safety:** Maintains proper separation between high-voltage and low-voltage cables, minimizing the risk of electromagnetic interference and electrical faults [9].

The 3D simulation diagram of cable laying effect using BIM technology is detailed in Figure 6

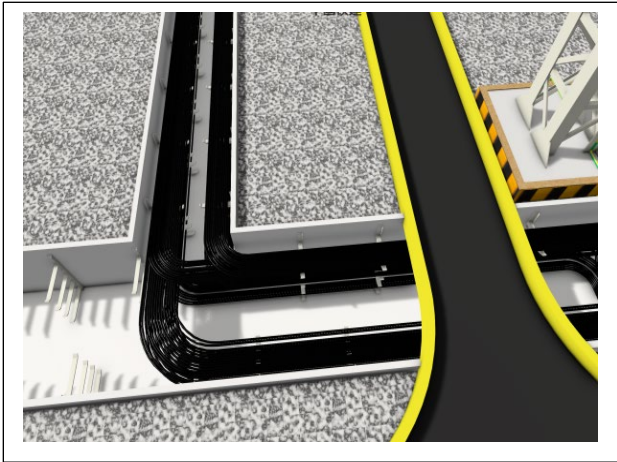


Fig. 6. Traction substation 3D cable routing

C. Visualization and Technical Briefing

Because BIM technology offers interactive simulations and immersive 3D visuals, it completely transforms the technical briefing process. Stakeholders can clearly grasp the design intent, construction timeline, and operating requirements with the help of process animation and virtual walkthroughs. The installation of secondary electrical systems was simulated using the BIM model, which highlighted important details, possible difficulties, and safety risks. By using these methods, construction teams can find and fix design errors before they appear on site, thereby enhancing construction quality and lowering the need for changes. Among the main benefits of BIM-based visualization are:

- **Improved Communication:** By providing a shared visual reference, BIM eliminates misunderstandings and ensures that all parties are aligned with the project's goals and requirements. This collaborative environment fosters better decision-making and streamlines the overall project workflow.
- **Enhanced Training:** BIM creates a virtual environment where operators can familiarize themselves with the substation layout, equipment placement, and operational procedures before the physical construction is completed.

- **Error Reduction:** potential issues such as spatial conflicts, accessibility challenges, or safety hazards can be detected early. This proactive error detection minimizes the likelihood of costly changes during construction, saving time, resources.

D. Application effect

Through the actual case verification, using Autodesk Revit as the core modeling tool, combined with 3dsMAX and Unity3D for visualization, BIM technology has shown significant advantages in collision detection, cable path optimization, construction simulation and other aspects [10]. Detailed data analysis is shown in Table 1.

TABLE I. BIM TECHNOLOGY APPLICATION DATA ANALYSIS

Metric	Application effect of BIM technology in traction substation		
	Traditional design	BIM technology	Improvement
Design cycle	16 weeks	12 weeks	Shorten by 25%
Collision inspection	10 collisions were detected manually	25 collisions were detected automatically	Increased by 150%
Cable path optimization	Cable waste 10.5%	Cable waste 5.5%	47.6% reduction of cable waste
Construction simulation accuracy	2D drawing with low accuracy	3D model with high accuracy	Increased by 60%
Total project cost	100%	90%	10%cost reduction

V. CONCLUSIONS

This study demonstrates the transformative potential of BIM technology in the design and management of traction substations. Through a detailed case study, the research highlights the methodology and outcomes of implementing BIM for traction substation modeling, encompassing data collection, equipment family library development, and the creation of comprehensive 3D models for the general layout and production house. The collaborative application of BIM technology has been thoroughly investigated, with a particular focus on its capabilities in collision detection, detailed design optimization, and visual representation. Key findings of this research include:

- **Enhanced Design Accuracy:** BIM technology significantly reduces design errors and omissions by providing a unified platform for interdisciplinary coordination and real-time clash detection.
- **Improved Construction Efficiency:** The use of 3D models for cable routing optimization and construction simulation minimizes rework, reduces material waste, and shortens project timelines.
- **Life-cycle Management:** BIM supports the entire lifecycle of traction substations, from design and construction to operation and maintenance, ensuring long-term efficiency and sustainability.

The adoption of BIM technology in traction substation design not only elevates the quality and efficiency of railway infrastructure projects but also accelerates the modernization and digital transformation of electrified railway systems. The

results of this study underscore the broad application prospects of BIM in the railway sector, paving the way for smarter, more resilient, and sustainable infrastructure development. Future research should explore the integration of BIM with emerging technologies such as artificial intelligence and the Internet of Things to further enhance the intelligence and automation of traction substation systems.

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